Shengjie Zhu

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About me

I'm an Applied Scientist at Amazon Ring AI. My Ph.D. research focused on 3D vision, including depth estimation, Structure-from-Motion, image correspondence esitmation, camera pose estimation, camera calibration, and etc.

Work Experience

Aug, 2024 – now **Applied Scientist II, Ring AI, Amazon.**

1. Developed *Ask Ring*, an agentic Ring AI assistant.
   - Led Intent Routing and Content Moderation development.
   - Contributed to Video Query Agent (VQA) development.
2. Optimized power- and memory-efficient Ring Animal Detection models for embedded devices.

May – Aug, 2024 **Research Scientist Intern, BAIR, Google.**
Research foundation depth model based Structure-from-Motion algorithms for Neural Rendering and AR/VR applications.

June – Sep, 2022 **Applied Scientist Intern, Amazon Device AI.**
Research SoTA few-shot object detection system.

June – Sep, 2021 **Applied Scientist Intern, Amazon Device AI.**
Research geometry algorithm to refine LiDAR depthmap, apply to driving DB including KITTI, Nuscenes, DDAD, Waymo.

Education

2017 – 2024: **PhD, Computer Science**, *Michigan State University*, East Lansing, U.S.
Dissertation: Structure-from-Motion from Dense Depth and Correspondence
Advisor: Prof. Xiaoming Liu
GPA: 3.70/4.0

2013 – 2017: **Bachelor of Engineering, Electrical and Electronics**, *Southeast University*, Nanjing, China.
GPA: 3.54/4.0

Publications

3DV'26 **Marginalized Bundle Adjustment: Multi-View Camera Pose from MonoDepth** [PDF, Code].
Shengjie Zhu, Ahmed Abdelkader, Mark J. Matthews, Xiaoming Liu, and Wen-Sheng Chu

ECCV'24 **Revisit Self-Supervision with Local Structure-from-Motion** [PDF, Code].
Shengjie Zhu, Xiaoming Liu

ECCV'24 **Remove Projective LiDAR Depthmap Artifacts via Exploiting Epipolar Geometry** [PDF, Code].
Shengjie Zhu*, Girish Chandar Ganesan*, Abhinav Kurmur, Xiaoming Liu

NeurIPS'23 **Tame a Wild Camera: In-the-Wild Monocular Camera Calibration** [PDF, Code].
Shengjie Zhu, Abhinav Kurmur, Masa Hu, Xiaoming Liu

CVPR'23 **LightedDepth: Video Depth Estimation in light of Limited Inference View Angles** [PDF, Code].
Shengjie Zhu, Xiaoming Liu

CVPR'23 **PMatch: Paired Masked Image Modeling for Dense Geometric Matching** [PDF, Code].
Shengjie Zhu, Xiaoming Liu

CVPR'20 **The Edge of Depth: Explicit Constraints between Segmentation and Depth** [PDF, Code].
Shengjie Zhu, Garrick Brazil, Xiaoming Liu

Computer Skills

Language **Python, C++, CUDA, Matlab, PyTorch, Tensorflow, CuPy, Numba**

Talk

Aug. 08, 2023 **3D Perception from Two Views**, Google Pixel Biometrics Seminar.
Feb. 05, 2024 **Structure-from-Motion Meets Self-supervised Learning**, CMU VASC Seminar. [Link]
Jul. 09, 2024 **Structure-from-Motion from Dense Learning**, Google Pixel Biometrics Seminar.